In chapter 8, I'll explain the retry pattern in detail. For now, I'll show you how to configure Catalog Service so that it retries connecting to Config Service a few times before failing (`spring.cloud.config.retry.max-attempts`). Each connection attempt is delayed according to an exponential backoff strategy, computed as the current delay multiplied by the value of the `spring.cloud.config.retry.multiplier` property. The initial delay is configured by `spring.cloud.config.retry.initial-interval`, and each delay cannot exceed the value of `spring.cloud.config.retry.max-interval`.

You can add the retry configuration to the application.yml file in the Catalog Service project.

Listing 4.18 Applying retry pattern to Spring Cloud Config Client

```
spring:
  application:
    name: catalog-service
  config:
    import: "optional:configserver:"
  cloud:
    config:
      uri: http://localhost:8888
      request-connect-timeout: 5000
      request-read-timeout: 5000
      fail-fast: true                ❶
      retry:
        max-attempts: 6              ❷
        initial-interval: 1000       ❸
        max-interval: 2000           ❹
        multiplier: 1.1              ❺
```

❶ Makes a failure to connect to the config server fatal

❷ Maximum number of attempts

❸ Initial retry interval for backoff (ms)

❹ Maximum retry interval for backoff (ms)